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Why Squatting Is the Anatomically Natural Defecation Posture for Humans

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Abstract
Motivation: 排便姿势是消化健康中一个重要却被忽视的因素。尽管西式坐便器在全球广泛使用，但进化解剖学与临床研究均表明，蹲姿才是更符合人体自然生理的最佳排便姿势。
Results: 通过综述解剖机制、临床研究、流行病学证据及进化学视角，本文表明：蹲姿可减少排便用力、缩短排便时间、降低肛肠压力，并降低常见肛肠及结肠疾病的发病风险。
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1. Introduction

Defecation is a basic physiological process shaped by millions of years of human evolution. For most of human history, humans eliminated waste in a deep squat position. Only in the past few centuries have sitting toilets become widespread in industrialized societies[1]. Today, a growing body of clinical and anatomical research indicates that the sitting posture misaligns the pelvic floor and creates unnecessary resistance during defecation. Excessive straining linked to sitting toilets has been associated with constipation, hemorrhoids, anal fissures, pelvic floor dysfunction, diverticulosis, and even cardiovascular strain during defecation. This paper synthesizes evidence from gastroenterology, anatomy, anthropology, and public health to establish that squatting is not merely a cultural preference but a biologically necessary posture for healthy defecation[2].

2.Results and discussions

2.1 Anatomical Mechanisms: How Squatting Unlocks the Rectum

The key anatomical structure governing defecation mechanics is the puborectalis muscle, a sling- like component of the pelvic floor that wraps around the rectum. At rest, this muscle maintains a slight contraction that creates a sharp bend — the anorectal angle — which preserves continence. During voluntary defecation, the muscle must relax to straighten this angle and open the passage for stool[3].

In a sitting posture, the anorectal angle typically remains near 90 – 100 degrees, maintaining a partial kink that resists stool passage. Users must strain forcibly to overcome this anatomical barrier. By contrast, squatting maximizes hip flexion, relaxes the puborectalis sling, and widens the anorectal angle to approximately 120 – 135 degrees, creating a nearly straight conduit from rectum to anus. This straightening eliminates the need for excessive pushing[4].

Squatting also increases intra - abdominal pressure gently and naturally through thigh - torso contact, supporting colonic motility without dangerous Valsalva straining. Together, these mechanisms make squatting the only posture that fully unlocks the body ’ s natural elimination system.Figure 1 shows the correct posture for defecation.

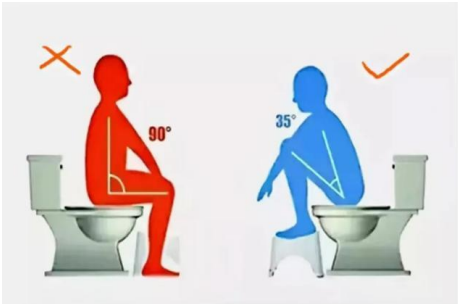

Fig1 The Proper Position for Defecation

2.2 Clinical Evidence: Squatting Improves Defecation Efficiency

Multiple controlled studies quantify the benefits of squatting over sitting. In a landmark study published in Digestive Diseases and Sciences, researchers measured defecation time and subjective effort across postures. Mean time to complete evacuation was 51 seconds in squatting versus 130 seconds in sitting. Participants reported significantly less straining and a stronger sense of complete emptying when squatting[5].

A study in the Journal of Clinical Gastroenterology confirmed that squatting improves rectal emptying and reduces residual stool. Another comparative trial found that squatting reduced intra-abdominal pressure spikes and lowered perineal descent, a risk factor for pelvic floor injury. These findings are reinforced by a scoping review in PMC that analyzed 42 studies: squatting consistently reduced straining, improved bowel clearance, and lowered discomfort compared to upright sitting.

Ergonomic modifications such as foot stools partially mimic squatting and reduce strain, but they cannot fully replicate the full hip flexion and pelvic alignment of a true squat.Figure 2 shows different types of shit.

Fig.2 How to Pass Healthy Stool

2.3 Epidemiological Links: Posture and Bowel Disease Risk

Cross- cultural and population- based studies suggest that defecation posture contributes to global variations in common bowel diseases. Diverticulosis—characterized by small pouches in the colon—is far more prevalent in populations that use sitting toilets. Chronic straining increases intraluminal colon pressure, which promotes herniation of the colonic wall.

Hemorrhoids and anal fissures are also closely tied to prolonged straining. By reducing pushing effort, squatting lowers venous pressure in the hemorrhoidal plexus and minimizes mucosal trauma. Studies comparing squat- using and sit- using populations show lower prevalence of hemorrhoidal disease where squatting is normative.

Squatting may also reduce the risk of defecation syncope (fainting) and cardiovascular events during elimination. Straining in the sitting posture increases intrathoracic pressure, reduces venous return, and can trigger dangerous blood- pressure fluctuations. The low- effort physiology of squatting reduces this cardiovascular load.Figure 3 illustrates the association between defecation posture and intestinal diseases.

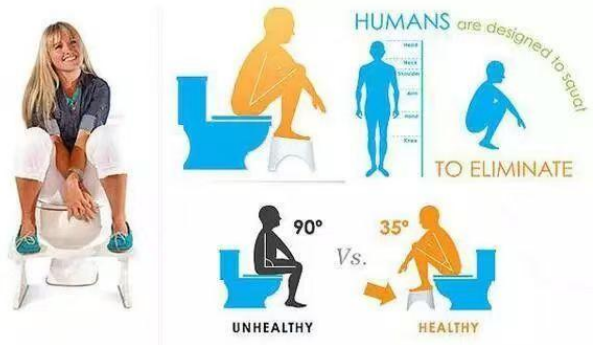

Fig 3: Association Between Defecation Posture and Intestinal Diseases

2.4 Evolutionary and Anthropological Perspective

Human anatomy evolved for squatting. The shape of the human pelvis, the angle of the femoral neck, the range of hip flexion, and the neuromuscular control of the pelvic floor all reflect adaptation to a deep- squat posture. Infants and young children naturally squat to eliminate, and traditional societies without modern toilets continue this practice[6].

Sitting toilets represent a cultural and engineering choice, not a biological improvement. Their rise coincided with urbanization and sanitation reforms but ignored basic pelvic anatomy. The global shift toward sitting has introduced a preventable layer of physiological strain into daily life.

2.5 Counterarguments and Special Populations

Critics note that squatting may challenge elderly individuals, people with mobility limitations, arthritis, or knee pain. For these groups, well- designed seated toilets with ergonomic foot supports provide a reasonable compromise. However, for most healthy adults, squatting is safe, sustainable, and far healthier than upright sitting.

Modern research also emphasizes that brief, functional squatting does not harm knees; it is prolonged static squatting that may cause fatigue. Defecation is a short activity, making the posture low- risk for most people.

3. Conclusion

The evidence is clear: squatting is the anatomically natural posture for human defecation. It straightens the anorectal angle, relaxes the puborectalis muscle, reduces straining, shortens evacuation time, improves colonic emptying, and lowers the risk of constipation, hemorrhoids, anal fissures, diverticulosis, and pelvic floor dysfunction. Evolution, anatomy, clinical trials, and population health all converge to show that sitting toilets disrupt normal physiology.

Public health education should promote squatting or semi- squatting ergonomics as a simple, non- invasive strategy to improve digestive health. Future toilet design should prioritize anatomical function over convention. For most people, returning to nature's design—squatting to defecate—is one of the most effective ways to support lifelong bowel health.

CRediT authorship contribution statement

Shangge Wang: Writing – review & editing, Visualization, Supervision, Conceptualization. Jiayou Cui: Writing – review & editing, Visualization, Supervision, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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